



The Valut 7 leaks

The leaked documents describing tools they use for spying, along with internal user manuals. <https://digit.in/jun20-118>



All about the NSO group

Everything we know about the NSO group and its links to various other companies and agencies. <https://digit.in/jun20-119>



SPY TOOLS

The covert technologies used by security agencies around the world

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ow does one come to know of the spy software used by intelligence agencies from around the

world? Most of the western software mentioned here was leaked by Edward Snowden, who exposed that the US, Australia, Canada, New Zealand and the United Kingdom were all involved in a collaborative network that intercepted communications between individuals around the world, and even on native soil. Specialised groups or think tanks that identify and track state actors over years have identified shady organisations and groups believed to be associated with certain Governments. Additionally, in some countries right to information requests have yielded some information. Apart from rare exceptions where a government declares that certain technologies are being used, most of the times it is never accepted that a certain software has been implemented and the extent to which it was used is typically downplayed. In case a country falls victim to an attack, or is breached in some way, such instances are also not likely to be reported or accepted. At times, such software have been discovered after botched attempts to install them. Note that for the most part, these names are the codenames, or what are officially known as covernames of the software.

Some of the publications to keep track of such surveillance campaigns and programs include The Intercept, Homeland Security Today and The Guardian. Another source of information are the blogs by cybersecurity firms such as Symantec and Kaspersky.

MASS SURVEILLANCE SOFTWARE

These are software that all work on the same basic principle. The physical infrastructure where most of the internet traffic in the world is funneled to, is tapped into. These may include national exchanges, international gateways, or fibre optic backbones. A simple way to understand this is that the signals are intercepted, and there are keyword based searches for certain words and phrases, such as “bomb” or “attack”. The monitoring is continuous and in real time. RAMPART-A is a US system disclosed by Snowden that began way back in 1992, when the fibre-optic infrastructure was being set up around the world. The global congestion points of fibre-optic traffic are tapped into, with agreements signed between participating nations. All kinds of communication are tapped by this system, including fax, VOIP calls and instant messages. RAMPART-A works in tandem with a XKeyscore, a program that can figure out the surfing habits of individual users, and create a profile known as a fingerprint.

After that, just based on surfing habits alone, a particular tagged user can be tracked in meatspace and cyberspace. These profiles allow agencies to bypass attempts to anonymise internet use. Tempora is a system used by the UK government which is similar to both RAMPART-A and XKeyscore. Perhaps the most well known mass surveillance software is PRISM. According to a leaked slideshow by Snowden, there are a number of big tech companies participating in the system, including Yahoo!, Google, Microsoft, Facebook and Apple. The system exploits the fact that most of the internet traffic passes through America, as the traffic has the tendency to use the cheapest available route rather than the most direct route. Most of the actual hardware for the internet is also located in the US. PRISM can apparently be used to intercept emails, voice chats, file transfers and social networking details.

India too has a system in place, called NETRA (Network TRaffic Analysis). That is the actual name of the system, and it has been publicly disclosed and discussed as well. The system is capable enough to monitor all kinds of things including status messages, video calls, tweets and shared images. Initially, the system was used by RAW, but is now used by domestic security agencies as well. The Russian version, called SORM, began by being